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Roll No. 311241 3042

324556(24)

BE (5th Semester)

Examination, Nov.-Dec., 2015

(New Scheme)

Communication Theory

Time Allowed : 3 hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each unit is compulsory. Attempt any two parts from (*b*), (*c*) and (*d*) in each Unit.

(*ii*) The figures in the right-hand margin indicate marks.

Unit - I

1. (*a*) What is need of modulation? 2

(*b*) Explain the phase-shift method of SSB generation with block diagram and suitable expression. 7

(*c*) Derive the expression for power content in AM wave and prove that

$$P_t = P_c\left(1+\frac{m_a^2}{2}\right)$$ 7

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(*d*) The antenna current of an AM transmitter is 8 A. If any carrier is sent it increases to 8.93 A. If carrier is modulated by a single sinusoidal wave, determine the percentage of modulation. Also find the antenna current if percentage of modulation changes to 80%. 7

Unit-II

2. (*a*) Derive relation between frequency and phase modulation. 2

(*b*) Explain pre-emphasis and de-emphasis circuit in detail. 7

(*c*) Describe reatance modulator method for FM generation. 7

(*d*) In an FM system, when the audio frequency is 500 Hz and the FM voltage is 2.4 volts, the deviation is 4.8 kHz. If AFM voltage is now increased to 7.2V, what is new deviation? Now, if the AFM voltage is raised to 10 V and AFM frequency is dropped to 200Hz, what is the deviation? Find the modulation index in each case. 7

Unit-III

3. (*a*) What is aperture effect? 2

(*b*) Explain generation and detection process of pulse width modulation (PWM) signal. 7

(*c*) A continuous time signal is given below : $x(t) = 8 \cos 200 \pi t$

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(*i*) Determine minimum sampling rate to avoid aliasing.

(*ii*) If sampling frequency f_s=400 Hz, what is the discrete time signal obtained after sampling?

(*iii*) If sampling frequency f_s=150 Hz, what is the discrete time signal obtained after sampling?

(*d*) Write shorts notes on : 7

(*i*) Time division multiplexing

(*ii*) Frequency division multiplexing

Unit-IV

4. (*a*) Explain the term 'quantization'. 2

(*b*) Derive an expression for signal to quantization noise ratio in PCM. 7

(*c*) Explain delta modulation in detail with suitable diagram. 7

(*d*) Explain generation and detection process of BPSK. 7

Unit-V

5. (*a*) What is entropy? 2

(*b*) The probabilities of the five possible symbols of a source are given as $P(x_1) = \frac{1}{2}$, $P(x_2) = \frac{1}{4}$, $P(x_3) = \frac{1}{8}$, $P(x_4) = P(x_5) = \frac{1}{16}$.

Determine the entropy and information rate if there are 16 symbols per second. 7

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(*c*) A DMS X has five equally likely symbols. Construct Shonnon-Fano code for X and calculate efficiency of the code. 7

(*d*) Write short notes on : 7

(*i*) Source coding

(*ii*) Codeword length

(*iii*) Code efficiency

(*iv*) Source coding theorem

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